Sharp Aggregate Bounds for Cluster-Induced Split Leakage: A Public ciFAIR Case Study

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Pioneer Research

***Abstract*—When record-level cluster memberships cannot be shared, an auditor may disclose only population size M, number of non-singleton groups G, their total membership S, and fixed test size T. Known-cluster hypergeometric formulas do not identify leakage because individual group sizes remain hidden. This paper proves sharp closed-form extrema for expected crossing groups and exposed test files over every compatible integer allocation. The crossing result holds for any nontrivial split; the exposure result allows groups larger than the test set when the test side is not the majority. A separate exact dynamic program checks the extrema. A public ciFAIR case study evaluates the calculations and target-definition sensitivity. For CIFAR-10, known-size expectations of 83.573 crossing groups and 86.097 exposed files lie inside aggregate-only bounds 80.722–84.446 and 85.390–86.668. CIFAR-100 gives 246.137 and 255.795 inside 231.559–248.615 and 252.226–257.504. The result is sharp partial identification of cluster-induced evaluation dependence, not a privacy guarantee, causal accuracy estimate, or duplicate detector.**

***Index Terms*—data leakage, duplicate clusters, partial identification, finite-population sampling, majorization**

I. Introduction

Train–test overlap can bias evaluation and destabilize model comparisons [1], [2]. Duplicate audits have changed conclusions for CIFAR and face datasets [3], [4], while recent visual audits scale retrieval to large benchmarks [5], [6]. Group-aware splitting is preferable when memberships are available, but restricted datasets may prohibit their disclosure.

This paper asks: *What split leakage is identifiable when only (M, G, S, T) may be published?* Its contributions are: (1) sharp expectation extrema over every hidden group-size allocation, including a broader parameter regime than the initial $K \leq T$ specialization; (2) an exact allocation dynamic program and pairwise-hypergeometric variance check for public known sizes; and (3) a reproducible ciFAIR case study with annotation-policy and single-edge-deletion sensitivity. The estimand is a counterfactual uniform fixed-size split, not classifier degradation.

II. Related Work

Guignard *et al.* derive multivariate-hypergeometric distributions and the known-size exposed-element expectation [7]; (2) below is its per-group specialization. The new object is the integer fiber $\{(k_1, \ldots, k_G) : k_i \geq 2, \sum k_i = S\}$ when the size vector is unavailable, and the contribution is sharp optimization of two functionals over that fiber. Classical occupancy majorization gives a conceptual antecedent [8], [9], but not these without-replacement aggregate bounds. DataSAIL instead constructs leakage-reduced splits from known similarities [10]. To the author's knowledge, the exact aggregate-only result has not appeared in the peer-reviewed cluster-leakage, splitting, or visual-duplicate literature searched through Aug. 3, 2026; the archived search log records databases, queries, and inclusion rules. This scoped claim does not assert a new majorization method.

III. Public Data and Operational Graphs

ciFAIR publishes CIFAR-10 and CIFAR-100 duplicate-pair annotations [3], [11]; CIFAR population and split sizes follow the primary dataset report [12]. At repository commit `4c2764277f5fda8fec6784a78c1818eab13236c5`, two CSVs per dataset list train–test and test–test pairs. Judgments denote approximate pixel copies (0), same-shot near-duplicates (1), and highly similar scenes (2).

Train and test identifiers use separate namespaces. The primary policy unions categories 0–2 and closes edges into connected components. Components of size at least two are operational dependence groups; unlisted files are residual records. Closure produces the disjoint groups required by the theorem, but similarity itself is not transitive and the components are not claimed as identities or ground-truth provenance classes. No image download or feature extraction is used. Both populations have $M = 60{,}000$ and $T = 10{,}000$ (Table I).

IV. Sharp Aggregate-Only Bounds

Let M, T, G, S be integers with $1 \leq T < M$, $G \geq 1$, and $2G \leq S \leq M$. Hidden group sizes $k_i \geq 2$ satisfy $\sum_i k_i = S$. Under a uniformly sampled test set of size T, a size-k group crosses train and test with probability

$$c(k) = 1 - \frac{\binom{M-k}{T}}{\binom{M}{T}} - \frac{\binom{M-k}{T-k}}{\binom{M}{T}}, \tag{1}$$

and its expected test files having a same-group training record are

$$e(k) = \frac{kT}{M} - k\frac{\binom{M-k}{T-k}}{\binom{M}{T}}. \tag{2}$$

TABLE I
PINNED ciFAIR CONSTRUCTION AND OFFICIAL-SPLIT OBSERVATION. HISTOGRAMS USE SIZE:COUNT.

Dataset	Edges	G	S	Max.	Group-size histogram	Crossing	Exposed
CIFAR-10	325	288	608	8	2:272, 3:8, 4:5, 6:2, 8:1	268	289
CIFAR-100	995	831	1,790	8	2:752, 3:51, 4:17, 5:4, 6:5, 7:1, 8:1	801	899

Out-of-range binomial coefficients are zero. Let $K = S - 2(G-1)$ be the largest feasible group.

Theorem 1. (a) For every $1 \le T < M$, $c(k)$ is nondecreasing and discrete-concave on $2 \le k \le K$. (b) If $T \le (M-1)/2$, $e(k)$ is strictly increasing and discrete-concave on the same domain; no $K \le T$ condition is needed. Consequently, either sum $\sum_i c(k_i)$ or $\sum_i e(k_i)$ is minimized by $(K, 2, \ldots, 2)$ and maximized by group sizes differing by at most one.

Proof. Write $(q)_k = q(q-1)\cdots(q-k+1)$, extended as zero for $k > q$, and define $a_k = (T)_k/(M)_k$, $b_k = (M-T)_k/(M)_k$, and $L = M - T$. For $u_q(k) = (q)_k/(M)_k$, direct subtraction gives

$$\Delta^2 u_q(k) = u_q(k)\frac{(M-q)(M-q-1)}{(M-k)(M-k-1)} \ge 0, \qquad (3)$$

including the boundary by zero extension. Hence $c(k) = 1 - a_k - b_k$ is concave. Adding a fixed group member under a coupled split cannot destroy a crossing, so c is nondecreasing.

For exposure and $k \le T$,

$$\Delta^2(ka_k) = a_k\frac{L[k(L+1)-2T]}{(M-k)(M-k-1)} > 0, \qquad (4)$$

because $L \ge T+1$ and $k \ge 2$; for $k > T$, $ka_k = 0$. Thus $e(k) = kT/M - ka_k$ is concave. Its differences equal $T/M > 0$ beyond T and, by concavity, are no smaller before T, proving strict increase. Finally, for concave f and $x \ge y+2$, balancing changes the sum by $\Delta f(y) - \Delta f(x-1) \ge 0$. Repetition yields the balanced maximum; reverse exchanges yield the concentrated minimum [9].

A separate exact check sets $E = S - 2G$, $D_0(0) = 0$, and

$$D_g(n) = \operatorname*{ext}_{0\le r\le n}\{D_{g-1}(n-r) + f(2+r)\}, \qquad (5)$$

with separate minima and maxima. This $O(GE^2)$ recurrence enumerates every excess-membership composition rather than relying on the closed-form witness.

V. PUBLIC CASE STUDY

Table II compares known-size expectations with bounds using only (M, G, S, T). Containment is a theorem consequence, so this is deterministic implementation validation, not an empirical hypothesis test. Exact pairwise inclusion–exclusion additionally gives crossing-count standard deviations of 7.617 and 12.915 for the known public histograms; variance is not claimed identifiable from the four aggregates. The crossing intervals span 4.46% and 6.93% of the known expectations, while exposure intervals span 1.48% and 2.06%.

Table III shows that annotation policy dominates hidden-size uncertainty: broadening the operational relation changes CIFAR-100 known crossing from 11.389 to 246.137. Category 0 has no CIFAR-10 edges; the displayed $[0, 0]$ interval is an explicit $G = S = 0$ convention outside Theorem 1. The result is therefore conditional on the declared graph policy, not robust to an unknown true duplicate relation. As a local perturbation check, deleting each published primary-policy edge in turn changes either known expectation by at most 0.278 for CIFAR-10 and 0.278 for CIFAR-100.

The official splits contain 268/801 crossing components and 289/899 exposed test files. Because the annotations and official split are coupled, these observations are descriptive and are not causal comparisons with the random-split model.

VI. REPRODUCIBILITY, RIGHTS, AND LIMITATIONS

The tagged public artifact is available at https://github.com/N-Mahesh/MOSAIC-EZ/tree/urtc2026-cifair-v2. It contains the PDF, source, frozen claim JSON, generated cell macros, manifest, code, search log, and tests. The verified environment is Python 3.12.5 with pypdf 6.14.2; the full check exits nonzero on checksum, schema, claim, PDF, manifest, or test failure and is expected to finish within two minutes. Table IV gives the tested command and exact upstream paths/hashes.

The pinned ciFAIR repository exposes no explicit license file for redistributing its metadata. The artifact therefore redistributes neither those CSVs nor CIFAR images; it stores hashes and derived aggregates and requires retrieval from the cited upstream source. Users remain responsible for upstream terms. No identifiable or new human-subject data are included.

The annotation graph omits a train–train census and is retrieval-biased. Missing or added edges may merge components, so neither G nor crossing count is a general lower bound; exposure is monotone only under added relations on a fixed record set. Category 2 and transitive closure are operational choices. Aggregate-only expectations do not identify variance, tail risk, classifier-score inflation, or realized leakage. Publishing (M, G, S, T) is disclosure-limited but provides no formal privacy guarantee.

VII. CONCLUSION

Discrete concavity yields sharp expectation intervals from (M, G, S, T) without memberships, over a wider regime than the initial small-group specialization. The public case study shows narrow allocation uncertainty but large annotation-policy sensitivity. These bounds can support a disclosure-limited audit when memberships cannot be published, but

TABLE II

KNOWN-SIZE VALUES VERSUS SHARP AGGREGATE-ONLY BOUNDS. PARENTHESES GIVE KNOWN-SIZE CROSSING SD.

Dataset	Known crossing (SD)	Crossing bounds	Known exposure	Exposure bounds
CIFAR-10	83.573 (7.617)	80.722–84.446	86.097	85.390–86.668
CIFAR-100	246.137 (12.915)	231.559–248.615	255.795	252.226–257.504

TABLE III

POLICY SENSITIVITY. EACH METRIC IS KNOWN VALUE / [AGGREGATE MINIMUM, MAXIMUM].

Dataset	Judgments	G	S	Crossing	Exposure
CIFAR-10	$\{0\}$	0	0	0.000/[0.000,0.000]	0.000/[0.000,0.000]
CIFAR-10	$\{0,1\}$	249	523	72.020/[69.883,72.640]	73.950/[73.390,74.376]
CIFAR-10	$\{0,1,2\}$	288	608	83.573/[80.722,84.446]	86.097/[85.390,86.668]
CIFAR-100	$\{0\}$	41	82	11.389/[11.389,11.389]	11.389/[11.389,11.389]
CIFAR-100	$\{0,1\}$	596	1,267	174.374/[166.281,175.975]	180.115/[178.114,181.183]
CIFAR-100	$\{0,1,2\}$	831	1,790	246.137/[231.559,248.615]	255.795/[252.226,257.504]

TABLE IV

COPY-PASTABLE ARTIFACT COMMAND AND PINNED UPSTREAM CIFAIR METADATA.

Command: `python tools/verify_public_release_bundle.py . --cifair-meta cifair/meta`	
`meta/duplicates_cifar10.csv`	`4cb7e99a7dfff346082c9d8fa4c2989a` `196e4a37d1e58f75936046696f1ba6a4`
`meta/duplicates_cifar10_test.csv`	`ef7b33e2f32d056cdb342046b437ee81` `9541a07cb3bb0520a3ada9b9c035f532`
`meta/duplicates_cifar100.csv`	`3891498a862f2d73df00cd5dc2ee9ae2` `b6dccf5238691d3c29d9abb12e1c63cb`
`meta/duplicates_cifar100_test.csv`	`78c9d8d17eda11b468a137119ed51f09` `332393d65a6329035e2250888edb2760`

group-aware splitting remains preferable and no causal accuracy or privacy claim follows.

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